TECHNICAL DOCUMENT

API Reference

docs/API_REFERENCE.md

REST · RFC 7807 · All Endpoints · All Services

PLATFORM	STACK	GENERATED
EduTech v1.0	Java 17 · Spring Boot 3.3.5	March 07, 2026

EduTech AI Platform — API Reference

Base URL: `https://api.edutech.com` (via api-gateway)
Authentication: `Authorization: Bearer <access_token>` on all endpoints except where noted
Content-Type: `application/json`
Error format: RFC 7807 ProblemDetail

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Common Conventions

Role Hierarchy

```
SUPER_ADMIN > CENTER_ADMIN > TEACHER > PARENT > STUDENT
```

Higher-privilege roles can perform all actions permitted to lower roles within their scope.

Common Response Codes

CODE	MEANING
200	OK
201	Created
202	Accepted (async operation initiated)
204	No Content
400	Validation error

CODE	MEANING
401	Missing or invalid JWT
403	Insufficient role
404	Resource not found
409	Conflict (duplicate resource)
422	Business rule violation
429	Rate limit exceeded
502	AI provider error (ai-gateway-svc only)

Error Body (RFC 7807 ProblemDetail)

All services return errors in the following format:

```
{
  "type": "https://edutech.com/problems/not-found",
  "title": "Not Found",
  "status": 404,
  "detail": "Exam with id 550e8400-e29b-41d4-a716-446655440000 not found"
}
```

Shared Object: DeviceFingerprint

Used in register, login, and refresh requests to bind sessions to specific devices.

FIELD	TYPE	REQUIRED	DESCRIPTION
`userAgent`	string	Yes	Browser or client user-agent string
`deviceId`	string	No	Client-generated stable device identifier
`ipSubnet`	string	No	IP subnet hint (e.g., `192.168.1.0/24`)

Authentication — auth-svc

Routes through api-gateway: `/api/v1/auth/**`, `/api/v1/otp/**`

POST /api/v1/auth/register

Register a new user account. Returns a token pair immediately on success; the account requires email verification before full access is granted.

Auth required: No

Request Body:

FIELD	TYPE	REQUIRED	CONSTRAINTS	DESCRIPTION
`email`	string	Yes	Valid email format	User's email address
`password`	string	Yes	8–128 characters	Account password
`role`	string (enum)	Yes	`SUPER_ADMIN`, `CENTER_ADMIN`, `TEACHER`, `PARENT`, `STUDENT`	Role to assign
`centerId`	UUID	No	—	Center to associate (required for non-SUPER_ADMIN roles)
`firstName`	string	Yes	Max 100 chars	User's first name
`lastName`	string	Yes	Max 100 chars	User's last name
`phoneNumber`	string	No	Max 20 chars	Contact phone number
`captchaToken`	string	Yes	—	CAPTCHA verification token
`deviceFingerprint`	DeviceFingerprint	Yes	See shared object	Device binding for refresh token

Response: `201 Created`

```
{
  "accessToken": "eyJhbGciOiJSUzI1NiJ9...",
  "refreshToken": "dGhpcyBpcyBhIHJlZnJlc2ggdG9rZW4...",
  "accessTokenExpiresAt": "2026-03-07T16:00:00Z",
  "refreshTokenExpiresAt": "2026-03-14T12:00:00Z",
  "tokenType": "Bearer"
}
```

Error Responses:

STATUS	CONDITION
400	Validation failure (invalid email, short password, etc.)
409	Email already registered
422	CAPTCHA verification failed

STATUS	CONDITION
429	Too many registration attempts

Example:

```
curl -X POST https://api.edutech.com/api/v1/auth/register \
  -H "Content-Type: application/json" \
  -d '{
    "email": "priya.sharma@example.com",
    "password": "Secure@1234",
    "role": "TEACHER",
    "centerId": "550e8400-e29b-41d4-a716-446655440000",
    "firstName": "Priya",
    "lastName": "Sharma",
    "phoneNumber": "+919876543210",
    "captchaToken": "03AGdBq25...",
    "deviceFingerprint": {
      "userAgent": "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7)",
      "deviceId": "device-abc-123",
      "ipSubnet": "203.0.113.0/24"
    }
  }'
```

POST /api/v1/auth/login

Authenticate an existing user and receive a JWT access/refresh token pair.

Auth required: No

Request Body:

FIELD	TYPE	REQUIRED	DESCRIPTION
`email`	string	Yes	Registered email address
`password`	string	Yes	Account password
`captchaToken`	string	Yes	CAPTCHA verification token
`deviceFingerprint`	DeviceFingerprint	Yes	Device binding for refresh token

Response: `200 OK`

```
{
  "accessToken": "eyJhbGciOiJSUzI1NiJ9...",
  "refreshToken": "dGhpcyBpcyBhIHJlZnJlc2ggdG9rZW4...",
```

```
    "accessTokenExpiresAt": "2026-03-07T16:00:00Z",
    "refreshTokenExpiresAt": "2026-03-14T12:00:00Z",
    "tokenType": "Bearer"
  }
```

Error Responses:

STATUS	CONDITION
400	Missing required fields
401	Invalid credentials
422	Account inactive or unverified
429	Rate limit exceeded (brute-force protection)

Example:

```
curl -X POST https://api.edutech.com/api/v1/auth/login \
  -H "Content-Type: application/json" \
  -d '{
    "email": "priya.sharma@example.com",
    "password": "Secure@1234",
    "captchaToken": "03AGdBq25...",
    "deviceFingerprint": {
      "userAgent": "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7)",
      "deviceId": "device-abc-123",
      "ipSubnet": "203.0.113.0/24"
    }
  }'
```

POST /api/v1/auth/refresh

Rotate the refresh token. Single-use and device-bound; the old refresh token is immediately invalidated.

Auth required: No

Request Body:

FIELD	TYPE	REQUIRED	DESCRIPTION
`refreshToken`	string	Yes	The current valid refresh token
`deviceFingerprint`	DeviceFingerprint	Yes	Must match the fingerprint used when the token was issued

Response: `200 OK` — same `TokenPair` structure as login.

Error Responses:

STATUS	CONDITION
400	Missing fields
401	Refresh token invalid, expired, or already used
403	Device fingerprint mismatch

Example:

```
curl -X POST https://api.edutech.com/api/v1/auth/refresh \
  -H "Content-Type: application/json" \
  -d '{
    "refreshToken": "dGhpcyBpcyBhIHJlZnJlc2ggdG9rZW4...",
    "deviceFingerprint": {
      "userAgent": "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7)",
      "deviceId": "device-abc-123",
      "ipSubnet": "203.0.113.0/24"
    }
  }'
```

POST /api/v1/auth/logout

Revoke the current session's refresh token.

Auth required: Yes (any role)

Request Body:

FIELD	TYPE	REQUIRED	DESCRIPTION
`refreshToken`	string	Yes	The refresh token to revoke
`deviceFingerprint`	DeviceFingerprint	Yes	Device fingerprint for validation

Response: `204 No Content`

Error Responses:

STATUS	CONDITION
401	Missing or invalid access token
404	Refresh token not found

Example:

```
curl -X POST https://api.edutech.com/api/v1/auth/logout \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..." \
  -H "Content-Type: application/json" \
  -d '{
    "refreshToken": "dGhpcyBpcyBhIHJlZnJlc2ggdG9rZW4...",
    "deviceFingerprint": {
      "userAgent": "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7)",
      "deviceId": "device-abc-123"
    }
  }'
```

POST /api/v1/auth/logout/all

Revoke all active refresh tokens for the authenticated user across all devices.

Auth required: Yes (any role)

Request Body: None

Response: `204 No Content`

Error Responses:

STATUS	CONDITION
401	Missing or invalid access token

Example:

```
curl -X POST https://api.edutech.com/api/v1/auth/logout/all \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

GET /api/v1/auth/me

Get the profile of the currently authenticated user.

Auth required: Yes (any role)

Response: `200 OK`

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "email": "priya.sharma@example.com",
  "firstName": "Priya",
  "lastName": "Sharma",
```

```
    "phoneNumber": "+919876543210",
    "role": "TEACHER",
    "status": "ACTIVE",
    "centerId": "661f9511-f3ac-52e5-b827-557766551111",
    "createdAt": "2026-01-15T10:30:00Z"
  }
```

Error Responses:

STATUS	CONDITION
401	Missing or invalid access token

Example:

```
curl https://api.edutech.com/api/v1/auth/me \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

POST /api/v1/otp/send

Send a one-time password to the specified email address via the chosen channel.

Auth required: No

Request Body:

FIELD	TYPE	REQUIRED	CONSTRAINTS	DESCRIPTION
`email`	string	Yes	Valid email	Recipient email address
`purpose`	string	Yes	—	OTP purpose (e.g., `EMAIL_VERIFICATION`, `PASSWORD_RESET`)
`channel`	string	Yes	`email` or `sms`	Delivery channel

Response: `202 Accepted` — No body.

Error Responses:

STATUS	CONDITION
400	Invalid email or channel value
404	Email not associated with any account
429	Too many OTP requests

Example:

```
curl -X POST https://api.edutech.com/api/v1/otp/send \
  -H "Content-Type: application/json" \
  -d '{
    "email": "priya.sharma@example.com",
    "purpose": "EMAIL_VERIFICATION",
    "channel": "email"
  }'
```

POST /api/v1/otp/verify

Verify a one-time password. If the purpose is `EMAIL_VERIFICATION`, the account is activated upon success.

Auth required: No

Request Body:

FIELD	TYPE	REQUIRED	CONSTRAINTS	DESCRIPTION
`email`	string	Yes	Valid email	Email for which OTP was sent
`otp`	string	Yes	Exactly 6 characters	The one-time password received
`purpose`	string	Yes	—	Must match the purpose used when sending

Response: `204 No Content`

Error Responses:

STATUS	CONDITION
400	Missing fields or OTP not exactly 6 characters
401	OTP incorrect or expired
404	No pending OTP found for this email and purpose

Example:

```
curl -X POST https://api.edutech.com/api/v1/otp/verify \
  -H "Content-Type: application/json" \
  -d '{
    "email": "priya.sharma@example.com",
    "otp": "482019",
    "purpose": "EMAIL_VERIFICATION"
  }'
```

POST /api/v1/auth/biometrics

Analyze keystroke dynamics from a login session and return a behavioral risk score. Scores range from 0.0 (low risk, consistent human typing) to 1.0 (high risk, bot-like behavior).

Auth required: Yes (any role)

Request Body:

FIELD	TYPE	REQUIRED	DESCRIPTION
`keystrokes`	array of KeystrokeEvent	Yes	Ordered list of key press events captured during typing
`sessionId`	string	Yes	Session identifier correlating keystroke data

KeystrokeEvent object:

FIELD	TYPE	DESCRIPTION
`keyDownTime`	long	Epoch milliseconds when the key was pressed
`keyUpTime`	long	Epoch milliseconds when the key was released
`key`	string	The key identifier (e.g., `"a"`, `"Shift"`)

Response: `200 OK`

```
{
  "score": 0.14,
  "level": "LOW",
  "sessionId": "sess-7f3a2b1c-4d5e-6f7a-8b9c-0d1e2f3a4b5c"
}
```

`level` is one of: `LOW` (score < 0.3), `MEDIUM` (0.3–0.7), `HIGH` (> 0.7).

Error Responses:

STATUS	CONDITION
400	Empty keystroke list or missing sessionId
401	Missing or invalid access token

Example:

```
curl -X POST https://api.edutech.com/api/v1/auth/biometrics \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..." \
```

```
  -H "Content-Type: application/json" \
  -d '{
    "sessionId": "sess-7f3a2b1c-4d5e-6f7a-8b9c-0d1e2f3a4b5c",
    "keystrokes": [
      { "keyDownTime": 1709808000000, "keyUpTime": 1709808000085, "key": "p" },
      { "keyDownTime": 1709808000140, "keyUpTime": 1709808000215, "key": "r" },
      { "keyDownTime": 1709808000270, "keyUpTime": 1709808000340, "key": "i" }
    ]
  }'
```

Coaching Centers — center-svc

Routes through api-gateway: `/api/v1/centers/**`

All endpoints require `Authorization: Bearer <access_token>`.

POST /api/v1/centers

Create a new coaching center. Only SUPER_ADMIN may perform this action.

Auth required: Yes — `SUPER_ADMIN`

Request Body:

FIELD	TYPE	REQUIRED	CONSTRAINTS	DESCRIPTION
`name`	string	Yes	Max 200 chars	Center display name
`code`	string	Yes	Max 20 chars, uppercase alphanumeric (`^[A-Z0-9]+$`)	Unique short code
`address`	string	Yes	Max 500 chars	Street address
`city`	string	Yes	Max 100 chars	City
`state`	string	Yes	Max 100 chars	State
`pincode`	string	Yes	Max 10 chars	Postal code
`phone`	string	Yes	Max 20 chars	Contact phone number
`email`	string	Yes	Valid email, max 255 chars	Contact email
`website`	string	No	Max 500 chars	Center website URL
`logoUrl`	string	No	Max 1000 chars	URL of the center's logo
`ownerId`	UUID	No	—	User ID to set as center owner

Response: `201 Created`

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "name": "Bright Futures Academy",
  "code": "BFA001",
  "address": "12 MG Road",
  "city": "Bengaluru",
  "state": "Karnataka",
  "pincode": "560001",
  "phone": "+918012345678",
  "email": "admin@brightfutures.edu",
  "website": "https://brightfutures.edu",
  "logoUrl": "https://cdn.edutech.com/logos/bfa001.png",
  "status": "ACTIVE",
  "ownerId": "661f9511-f3ac-52e5-b827-557766551111",
  "createdAt": "2026-03-07T10:00:00Z",
  "updatedAt": "2026-03-07T10:00:00Z"
}
```

Error Responses:

STATUS	CONDITION
400	Validation failure
403	Caller is not SUPER_ADMIN
409	Center code already exists

Example:

```
curl -X POST https://api.edutech.com/api/v1/centers \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..." \
  -H "Content-Type: application/json" \
  -d '{
    "name": "Bright Futures Academy",
    "code": "BFA001",
    "address": "12 MG Road",
    "city": "Bengaluru",
    "state": "Karnataka",
    "pincode": "560001",
    "phone": "+918012345678",
    "email": "admin@brightfutures.edu",
    "website": "https://brightfutures.edu",
    "ownerId": "661f9511-f3ac-52e5-b827-557766551111"
  }'
```

GET /api/v1/centers

List coaching centers accessible to the authenticated caller. SUPER_ADMIN sees all centers; CENTER_ADMIN and below see only their own center.

Auth required: Yes (any role)

Response: `200 OK` — Array of CenterResponse objects (see POST response schema).

Example:

```
curl https://api.edutech.com/api/v1/centers \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

GET /api/v1/centers/{centerId}

Get full details of a specific coaching center.

Auth required: Yes (any role with center access)

Path Parameters:

NAME	TYPE	DESCRIPTION
`centerId`	UUID	Center identifier

Response: `200 OK` — CenterResponse object (see POST response schema).

Error Responses:

STATUS	CONDITION
403	Caller does not have access to this center
404	Center not found

Example:

```
curl https://api.edutech.com/api/v1/centers/550e8400-e29b-41d4-a716-446655440000 \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

PUT /api/v1/centers/{centerId}

Update mutable details of a coaching center.

Auth required: Yes — `SUPER_ADMIN` or `CENTER_ADMIN` of this center

Path Parameters:

NAME	TYPE	DESCRIPTION
`centerId`	UUID	Center identifier

Request Body:

FIELD	TYPE	REQUIRED	CONSTRAINTS	DESCRIPTION
`name`	string	Yes	Max 200 chars	Center display name
`address`	string	Yes	Max 500 chars	Street address
`city`	string	Yes	Max 100 chars	City
`state`	string	Yes	Max 100 chars	State
`pincode`	string	Yes	Max 10 chars	Postal code
`phone`	string	Yes	Max 20 chars	Contact phone number
`email`	string	Yes	Valid email, max 255 chars	Contact email
`website`	string	No	Max 500 chars	Center website URL
`logoUrl`	string	No	Max 1000 chars	URL of center logo

Response: `200 OK` — Updated CenterResponse object.

Error Responses:

STATUS	CONDITION
400	Validation failure
403	Insufficient role
404	Center not found

Example:

```
curl -X PUT https://api.edutech.com/api/v1/centers/550e8400-e29b-41d4-a716-446655440000 \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..." \
  -H "Content-Type: application/json" \
  -d '{
    "name": "Bright Futures Academy — Koramangala",
    "address": "45 Sarjapur Road",
    "city": "Bengaluru",
    "state": "Karnataka",
    "pincode": "560034",
```

```
  "phone": "+918012345678",
  "email": "koramangala@brightfutures.edu"
}'
```

POST /api/v1/centers/{centerId}/batches

Create a new batch within a center.

Auth required: Yes — `CENTER_ADMIN` or `SUPER_ADMIN`

Path Parameters:

NAME	TYPE	DESCRIPTION
`centerId`	UUID	Center identifier

Request Body:

FIELD	TYPE	REQUIRED	CONSTRAINTS	DESCRIPTION
`name`	string	Yes	Max 200 chars	Batch name
`code`	string	Yes	Max 50 chars	Unique batch code within the center
`subject`	string	Yes	Max 100 chars	Subject or course name
`teacherId`	UUID	No	—	Assigned teacher's user ID
`maxStudents`	int	Yes	1–200	Maximum student capacity
`startDate`	date	Yes	ISO-8601 (`YYYY-MM-DD`)	Batch start date
`endDate`	date	No	ISO-8601 (`YYYY-MM-DD`)	Batch end date

Response: `201 Created`

```
{
  "id": "772g0622-g4bd-63f6-c938-668877662222",
  "centerId": "550e8400-e29b-41d4-a716-446655440000",
  "name": "JEE Advanced 2027 — Batch A",
  "code": "JEE-A-2027",
  "subject": "Physics, Chemistry, Mathematics",
  "teacherId": "661f9511-f3ac-52e5-b827-557766551111",
  "maxStudents": 40,
  "enrolledCount": 0,
  "startDate": "2026-06-01",
  "endDate": "2027-04-30",
  "status": "ACTIVE",
```

```
    "createdAt": "2026-03-07T10:00:00Z",
    "updatedAt": "2026-03-07T10:00:00Z"
  }
```

Error Responses:

STATUS	CONDITION
400	Validation failure
403	Insufficient role
404	Center not found
409	Batch code already exists in this center

Example:

```
curl -X POST https://api.edutech.com/api/v1/centers/550e8400-e29b-41d4-a716-446655440000/batches \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..." \
  -H "Content-Type: application/json" \
  -d '{
    "name": "JEE Advanced 2027 — Batch A",
    "code": "JEE-A-2027",
    "subject": "Physics, Chemistry, Mathematics",
    "teacherId": "661f9511-f3ac-52e5-b827-557766551111",
    "maxStudents": 40,
    "startDate": "2026-06-01",
    "endDate": "2027-04-30"
  }'
```

GET /api/v1/centers/{centerId}/batches

List batches for a center, optionally filtered by status.

Auth required: Yes (any role with center access)

Path Parameters:

NAME	TYPE	DESCRIPTION
`centerId`	UUID	Center identifier

Query Parameters:

NAME	TYPE	REQUIRED	DESCRIPTION
`status`	string (BatchStatus enum)	No	Filter by status (e.g., `ACTIVE`, `COMPLETED`, `CANCELLED`)

Response: `200 OK` — Array of BatchResponse objects.

Example:

```
curl "https://api.edutech.com/api/v1/centers/550e8400-e29b-41d4-a716-446655440000/batches?status=ACTIVE" \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

GET /api/v1/centers/{centerId}/batches/{batchId}

Get details of a specific batch.

Auth required: Yes (any role with center access)

Path Parameters:

NAME	TYPE	DESCRIPTION
`centerId`	UUID	Center identifier
`batchId`	UUID	Batch identifier

Response: `200 OK` — BatchResponse object (see POST /batches response schema).

Error Responses:

STATUS	CONDITION
404	Center or batch not found
403	Insufficient access

Example:

```
curl https://api.edutech.com/api/v1/centers/550e8400-e29b-41d4-a716-446655440000/batches/772g0622-g4bd-63f6-c938-668877662222 \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

PUT /api/v1/centers/{centerId}/batches/{batchId}

Update a batch's status or teacher assignment.

Auth required: Yes — `CENTER_ADMIN` or `SUPER_ADMIN`

Path Parameters:

NAME	TYPE	DESCRIPTION
`centerId`	UUID	Center identifier
`batchId`	UUID	Batch identifier

Request Body:

FIELD	TYPE	REQUIRED	DESCRIPTION
`teacherId`	UUID	No	New teacher assignment (null removes the assignment)
`status`	string (BatchStatus)	Yes	Updated batch status

Response: `200 OK` — Updated BatchResponse object.

Error Responses:

STATUS	CONDITION
400	Missing required status field
403	Insufficient role
404	Batch not found

Example:

```
curl -X PUT https://api.edutech.com/api/v1/centers/550e8400-e29b-41d4-a716-446655440000/batches/772g0622-g4bd-63f6-c938-668877662222 \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..." \
  -H "Content-Type: application/json" \
  -d '{
    "teacherId": "883h1733-h5ce-74g7-d049-779988773333",
    "status": "ACTIVE"
  }'
```

POST /api/v1/centers/{centerId}/teachers

Assign a user as a teacher to the specified center.

Auth required: Yes — `CENTER_ADMIN` or `SUPER_ADMIN`

Path Parameters:

NAME	TYPE	DESCRIPTION
`centerId`	UUID	Center identifier

Request Body:

FIELD	TYPE	REQUIRED	CONSTRAINTS	DESCRIPTION
`userId`	UUID	Yes	—	Auth-svc user ID to assign as teacher
`firstName`	string	Yes	Max 100 chars	Teacher's first name
`lastName`	string	Yes	Max 100 chars	Teacher's last name
`email`	string	Yes	Valid email, max 255 chars	Teacher's email
`phoneNumber`	string	No	Max 20 chars	Teacher's phone number
`subjects`	string	No	Max 500 chars	Comma-separated list of subjects taught

Response: `201 Created`

```
{
  "id": "994i2844-i6df-85h8-e150-880099884444",
  "centerId": "550e8400-e29b-41d4-a716-446655440000",
  "userId": "661f9511-f3ac-52e5-b827-557766551111",
  "firstName": "Priya",
  "lastName": "Sharma",
  "email": "priya.sharma@example.com",
  "phoneNumber": "+919876543210",
  "subjects": "Physics, Mathematics",
  "status": "ACTIVE",
  "joinedAt": "2026-03-07T10:00:00Z"
}
```

Error Responses:

STATUS	CONDITION
400	Validation failure
403	Insufficient role
404	Center or user not found
409	User already assigned to this center

Example:

```
curl -X POST https://api.edutech.com/api/v1/centers/550e8400-e29b-41d4-a716-446655440000/teachers \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..." \
  -H "Content-Type: application/json" \
  -d '{
    "userId": "661f9511-f3ac-52e5-b827-557766551111",
    "firstName": "Priya",
    "lastName": "Sharma",
    "email": "priya.sharma@example.com",
    "phoneNumber": "+919876543210",
    "subjects": "Physics, Mathematics"
  }'
```

GET /api/v1/centers/{centerId}/teachers

List all teachers assigned to a center.

Auth required: Yes (any role with center access)

Path Parameters:

NAME	TYPE	DESCRIPTION
`centerId`	UUID	Center identifier

Response: `200 OK` — Array of TeacherResponse objects (see POST /teachers response schema).

Example:

```
curl https://api.edutech.com/api/v1/centers/550e8400-e29b-41d4-a716-446655440000/teachers \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

POST /api/v1/centers/{centerId}/batches/{batchId}/schedules

Add a recurring schedule slot to a batch.

Auth required: Yes — `CENTER_ADMIN`, `TEACHER`, or `SUPER_ADMIN`

Path Parameters:

NAME	TYPE	DESCRIPTION
`centerId`	UUID	Center identifier
`batchId`	UUID	Batch identifier

Request Body:

FIELD	TYPE	REQUIRED	DESCRIPTION
`dayOfWeek`	string	Yes	Day of week (`MONDAY`, `TUESDAY`, ..., `SUNDAY`)
`startTime`	string	Yes	ISO-8601 local time (`HH:mm:ss`)
`endTime`	string	Yes	ISO-8601 local time (`HH:mm:ss`)
`room`	string	Yes	Max 100 chars — Room or venue name
`effectiveFrom`	date	Yes	ISO-8601 date (`YYYY-MM-DD`) — Schedule start date
`effectiveTo`	date	No	ISO-8601 date (`YYYY-MM-DD`) — Schedule end date

Response: `201 Created`

```
{
  "id": "aa5j3955-j7eg-96i9-f261-991100995555",
  "batchId": "772g0622-g4bd-63f6-c938-668877662222",
  "centerId": "550e8400-e29b-41d4-a716-446655440000",
  "dayOfWeek": "MONDAY",
  "startTime": "09:00:00",
  "endTime": "11:00:00",
  "room": "Lab 3",
  "effectiveFrom": "2026-06-01",
  "effectiveTo": "2027-04-30",
  "createdAt": "2026-03-07T10:00:00Z"
}
```

Error Responses:

STATUS	CONDITION
400	Invalid time format or missing fields
403	Insufficient role
404	Batch or center not found
409	Room conflict for the same time slot

Example:

```
curl -X POST "https://api.edutech.com/api/v1/centers/550e8400-e29b-41d4-a716-446655440000/batches/
772g0622-g4bd-63f6-c938-668877662222/schedules" \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..." \
  -H "Content-Type: application/json" \
```

```
  -d '{
    "dayOfWeek": "MONDAY",
    "startTime": "09:00:00",
    "endTime": "11:00:00",
    "room": "Lab 3",
    "effectiveFrom": "2026-06-01",
    "effectiveTo": "2027-04-30"
  }'
```

GET /api/v1/centers/{centerId}/batches/{batchId}/schedules

List all schedule slots for a batch.

Auth required: Yes (any role with center access)

Path Parameters:

NAME	TYPE	DESCRIPTION
`centerId`	UUID	Center identifier
`batchId`	UUID	Batch identifier

Response: `200 OK` — Array of ScheduleResponse objects.

Example:

```
curl "https://api.edutech.com/api/v1/centers/550e8400-e29b-41d4-a716-446655440000/batches/772g0622
-g4bd-63f6-c938-668877662222/schedules" \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

POST /api/v1/centers/{centerId}/fees

Create a fee structure for a center.

Auth required: Yes — `CENTER_ADMIN` or `SUPER_ADMIN`

Path Parameters:

NAME	TYPE	DESCRIPTION
`centerId`	UUID	Center identifier

Request Body:

FIELD	TYPE	REQUIRED	CONSTRAINTS	DESCRIPTION
`name`	string	Yes	Max 200 chars	Fee structure name
`description`	string	No	Max 1000 chars	Description of what is included
`amount`	decimal	Yes	Min 0.01	Fee amount
`currency`	string	Yes	Max 5 chars (e.g., `INR`, `USD`)	Currency code
`frequency`	string (FeeFrequency enum)	Yes	—	Billing frequency (e.g., `MONTHLY`, `QUARTERLY`, `ANNUALLY`)
`dueDay`	int	Yes	1–31	Day of month when payment is due
`lateFeeAmount`	decimal	No	Min 0.00	Late payment penalty amount

Response: `201 Created`

```
{
  "id": "bb6k4066-k8fh-07j0-g372-002211006666",
  "centerId": "550e8400-e29b-41d4-a716-446655440000",
  "name": "JEE Standard Monthly Plan",
  "description": "Monthly fee for JEE Advanced batch",
  "amount": 5000.00,
  "currency": "INR",
  "frequency": "MONTHLY",
  "dueDay": 5,
  "lateFeeAmount": 250.00,
  "status": "ACTIVE",
  "createdAt": "2026-03-07T10:00:00Z"
}
```

Error Responses:

STATUS	CONDITION
400	Validation failure
403	Insufficient role
404	Center not found

Example:

```
curl -X POST https://api.edutech.com/api/v1/centers/550e8400-e29b-41d4-a716-446655440000/fees \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..." \
  -H "Content-Type: application/json" \
  -d '{
    "name": "JEE Standard Monthly Plan",
    "description": "Monthly fee for JEE Advanced batch",
    "amount": 5000.00,
    "currency": "INR",
    "frequency": "MONTHLY",
    "dueDay": 5,
    "lateFeeAmount": 250.00
  }'
```

GET /api/v1/centers/{centerId}/fees

List active fee structures for a center.

Auth required: Yes (any role with center access)

Path Parameters:

NAME	TYPE	DESCRIPTION
`centerId`	UUID	Center identifier

Response: `200 OK` — Array of FeeStructureResponse objects.

Example:

```
curl https://api.edutech.com/api/v1/centers/550e8400-e29b-41d4-a716-446655440000/fees \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

POST /api/v1/centers/{centerId}/batches/{batchId}/attendance

Mark or re-mark attendance for an entire batch on a specific date. Re-posting for the same date replaces existing records.

Auth required: Yes — `TEACHER`, `CENTER_ADMIN`, or `SUPER_ADMIN`

Path Parameters:

NAME	TYPE	DESCRIPTION
`centerId`	UUID	Center identifier

NAME	TYPE	DESCRIPTION
`batchId`	UUID	Batch identifier

Request Body:

FIELD	TYPE	REQUIRED	DESCRIPTION
`date`	date	Yes	ISO-8601 date (`YYYY-MM-DD`) — Attendance date
`entries`	array of AttendanceEntry	Yes	One entry per student

AttendanceEntry object:

FIELD	TYPE	REQUIRED	DESCRIPTION
`studentId`	UUID	Yes	Student's user ID
`status`	string (AttendanceStatus enum)	Yes	`PRESENT`, `ABSENT`, `LATE`, or `EXCUSED`
`notes`	string	No	Optional note for this student

Response: `201 Created` — Array of AttendanceResponse objects.

```
[
  {
    "id": "cc715177-19gi-18k1-h483-113322117777",
    "batchId": "772g0622-g4bd-63f6-c938-668877662222",
    "studentId": "dd8m6288-m0hj-29l2-i594-224433228888",
    "date": "2026-03-07",
    "status": "PRESENT",
    "markedByTeacherId": "994i2844-i6df-85h8-e150-880099884444",
    "notes": null,
    "createdAt": "2026-03-07T09:05:00Z"
  }
]
```

Error Responses:

STATUS	CONDITION
400	Invalid date format or empty entries list
403	Insufficient role
404	Batch or student not found

Example:

```
curl -X POST "https://api.edutech.com/api/v1/centers/550e8400-e29b-41d4-a716-446655440000/batches/
772g0622-g4bd-63f6-c938-668877662222/attendance" \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..." \
  -H "Content-Type: application/json" \
  -d '{
    "date": "2026-03-07",
    "entries": [
      { "studentId": "dd8m6288-m0hj-2912-i594-224433228888", "status": "PRESENT" },
      { "studentId": "ee9n7399-n1ik-30m3-j605-335544339999", "status": "ABSENT", "notes": "Called
in sick" }
    ]
  }'
```

GET /api/v1/centers/{centerId}/batches/{batchId}/attendance

Retrieve attendance records for a batch on a specific date.

Auth required: Yes (any role with center access)

Path Parameters:

NAME	TYPE	DESCRIPTION
`centerId`	UUID	Center identifier
`batchId`	UUID	Batch identifier

Query Parameters:

NAME	TYPE	REQUIRED	DESCRIPTION
`date`	date	Yes	ISO-8601 date (`YYYY-MM-DD`) — Date to query

Response: `200 OK` — Array of AttendanceResponse objects.

Error Responses:

STATUS	CONDITION
400	Missing or invalid date format
404	Batch not found

Example:

```
curl "https://api.edutech.com/api/v1/centers/550e8400-e29b-41d4-a716-446655440000/batches/772g0622
-g4bd-63f6-c938-668877662222/attendance?date=2026-03-07" \
```

```
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

POST /api/v1/centers/{centerId}/content

Register a content item's metadata. The file must already be uploaded to CDN or S3 before calling this endpoint.

Auth required: Yes — `TEACHER`, `CENTER_ADMIN`, or `SUPER_ADMIN`

Path Parameters:

NAME	TYPE	DESCRIPTION
`centerId`	UUID	Center identifier

Request Body:

FIELD	TYPE	REQUIRED	CONSTRAINTS	DESCRIPTION
`batchId`	UUID	No	—	Batch this content belongs to (null = center-wide)
`title`	string	Yes	Max 500 chars	Content title
`description`	string	No	Max 2000 chars	Content description
`type`	string (ContentType enum)	Yes	—	Type (e.g., `PDF`, `VIDEO`, `AUDIO`, `IMAGE`, `DOCUMENT`)
`fileUrl`	string	Yes	Max 2000 chars	CDN/S3 URL of the uploaded file
`fileSizeBytes`	long	No	Min 0	File size in bytes

Response: `201 Created`

```
{
  "id": "ff0o8400-o2j1-41n4-k716-446655440000",
  "centerId": "550e8400-e29b-41d4-a716-446655440000",
  "batchId": "772g0622-g4bd-63f6-c938-668877662222",
  "title": "Chapter 3 — Kinematics Notes",
  "description": "Detailed handwritten notes on projectile motion",
  "type": "PDF",
  "fileUrl": "https://cdn.edutech.com/content/ch3-kinematics.pdf",
  "fileSizeBytes": 2048000,
  "uploadedByUserId": "994i2844-i6df-85h8-e150-880099884444",
  "status": "ACTIVE",
```

```
  "createdAt": "2026-03-07T11:00:00Z"
}
```

Error Responses:

STATUS	CONDITION
400	Validation failure
403	Insufficient role
404	Center not found

Example:

```
curl -X POST https://api.edutech.com/api/v1/centers/550e8400-e29b-41d4-a716-446655440000/content \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..." \
  -H "Content-Type: application/json" \
  -d '{
    "batchId": "772g0622-g4bd-63f6-c938-668877662222",
    "title": "Chapter 3 — Kinematics Notes",
    "description": "Detailed handwritten notes on projectile motion",
    "type": "PDF",
    "fileUrl": "https://cdn.edutech.com/content/ch3-kinematics.pdf",
    "fileSizeBytes": 2048000
  }'
```

GET /api/v1/centers/{centerId}/content

List all content items for a center.

Auth required: Yes (any role with center access)

Path Parameters:

NAME	TYPE	DESCRIPTION
`centerId`	UUID	Center identifier

Response: `200 OK` — Array of ContentItemResponse objects.

Example:

```
curl https://api.edutech.com/api/v1/centers/550e8400-e29b-41d4-a716-446655440000/content \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

Parent Portal — parent-svc

Routes through api-gateway: `/api/v1/parents/**`

All endpoints require `Authorization: Bearer <access_token>`.

POST /api/v1/parents

Create a parent profile for the authenticated user. A user may only have one parent profile.

Auth required: Yes — `PARENT` role

Request Body:

FIELD	TYPE	REQUIRED	CONSTRAINTS	DESCRIPTION
`name`	string	Yes	Non-blank	Parent's display name
`phone`	string	No	Pattern `^\+?[0-9]{7,15}$`	Contact phone number

Response: `201 Created`

```
{
  "id": "gg1p9511-p3km-52o5-1827-557766551111",
  "userId": "550e8400-e29b-41d4-a716-446655440000",
  "name": "Ramesh Kumar",
  "phone": "+919123456789",
  "verified": false,
  "status": "ACTIVE",
  "createdAt": "2026-03-07T10:00:00Z"
}
```

Error Responses:

STATUS	CONDITION
400	Invalid phone number format
403	Caller does not have PARENT role
409	Profile already exists for this user

Example:

```
curl -X POST https://api.edutech.com/api/v1/parents \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..." \
  -H "Content-Type: application/json" \
  -d '{
```

```
    "name": "Ramesh Kumar",
    "phone": "+919123456789"
  }'
```

GET /api/v1/parents/me

Get the authenticated parent's own profile.

Auth required: Yes — `PARENT` role

Response: `200 OK` — ParentProfileResponse (see POST /parents response schema).

Error Responses:

STATUS	CONDITION
404	Profile not yet created for this user

Example:

```
curl https://api.edutech.com/api/v1/parents/me \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

GET /api/v1/parents/{profileId}

Get a parent profile by its ID. Accessible by SUPER_ADMIN and CENTER_ADMIN; parents may only access their own profile.

Auth required: Yes — `PARENT` (own profile only), `CENTER_ADMIN`, or `SUPER_ADMIN`

Path Parameters:

NAME	TYPE	DESCRIPTION
`profileId`	UUID	Parent profile identifier

Response: `200 OK` — ParentProfileResponse.

Error Responses:

STATUS	CONDITION
403	Caller attempting to access another parent's profile
404	Profile not found

Example:

```
curl https://api.edutech.com/api/v1/parents/gg1p9511-p3km-52o5-1827-557766551111 \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

PUT /api/v1/parents/{profileId}

Update a parent profile's name and/or phone number.

Auth required: Yes — profile owner or `SUPER_ADMIN`

Path Parameters:

NAME	TYPE	DESCRIPTION
`profileId`	UUID	Parent profile identifier

Request Body:

FIELD	TYPE	REQUIRED	DESCRIPTION
`name`	string	No	Updated display name
`phone`	string	No	Updated phone number

Response: `200 OK` — Updated ParentProfileResponse.

Error Responses:

STATUS	CONDITION
403	Not the profile owner
404	Profile not found

Example:

```
curl -X PUT https://api.edutech.com/api/v1/parents/gg1p9511-p3km-52o5-1827-557766551111 \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..." \
  -H "Content-Type: application/json" \
  -d '{ "name": "Ramesh Kumar Pillai", "phone": "+919900001234" }'
```

POST /api/v1/parents/{profileId}/students

Link a student to this parent profile.

Auth required: Yes — profile owner or `CENTER_ADMIN` / `SUPER_ADMIN`

Path Parameters:

NAME	TYPE	DESCRIPTION
`profileId`	UUID	Parent profile identifier

Request Body:

FIELD	TYPE	REQUIRED	DESCRIPTION
`studentId`	UUID	Yes	Auth-svc user ID of the student
`studentName`	string	Yes	Student's display name
`centerId`	UUID	Yes	Center where the student is enrolled

Response: `201 Created`

```
{
  "id": "hh2q0622-q4ln-63p6-m938-668877662222",
  "parentId": "gg1p9511-p3km-52o5-l827-557766551111",
  "studentId": "dd8m6288-m0hj-2912-i594-224433228888",
  "studentName": "Ananya Kumar",
  "centerId": "550e8400-e29b-41d4-a716-446655440000",
  "status": "ACTIVE",
  "createdAt": "2026-03-07T10:00:00Z"
}
```

Error Responses:

STATUS	CONDITION
400	Missing required fields
403	Insufficient access
404	Parent profile or student not found
409	Student already linked to this parent

Example:

```
curl -X POST https://api.edutech.com/api/v1/parents/gg1p9511-p3km-52o5-l827-557766551111/students
\
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..." \
  -H "Content-Type: application/json" \
  -d '{
    "studentId": "dd8m6288-m0hj-2912-i594-224433228888",
```

```
    "studentName": "Ananya Kumar",
    "centerId": "550e8400-e29b-41d4-a716-446655440000"
  }'
```

GET /api/v1/parents/{profileId}/students

List all active student links for a parent profile.

Auth required: Yes — profile owner or `CENTER_ADMIN` / `SUPER_ADMIN`

Path Parameters:

NAME	TYPE	DESCRIPTION
`profileId`	UUID	Parent profile identifier

Response: `200 OK` — Array of StudentLinkResponse objects.

Example:

```
curl https://api.edutech.com/api/v1/parents/gg1p9511-p3km-52o5-1827-557766551111/students \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

DELETE /api/v1/parents/{profileId}/students/{linkId}

Revoke a student link from a parent profile.

Auth required: Yes — profile owner or `CENTER_ADMIN` / `SUPER_ADMIN`

Path Parameters:

NAME	TYPE	DESCRIPTION
`profileId`	UUID	Parent profile identifier
`linkId`	UUID	Student link identifier

Response: `204 No Content`

Error Responses:

STATUS	CONDITION
403	Not the profile owner
404	Link not found

Example:

```
curl -X DELETE https://api.edutech.com/api/v1/parents/gg1p9511-p3km-52o5-1827-557766551111/students/hh2q0622-q4ln-63p6-m938-668877662222 \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

POST /api/v1/parents/{profileId}/payments

Record a fee payment for a student.

Auth required: Yes — profile owner or `CENTER_ADMIN` / `SUPER_ADMIN`

Path Parameters:

NAME	TYPE	DESCRIPTION
`profileId`	UUID	Parent profile identifier

Request Body:

FIELD	TYPE	REQUIRED	CONSTRAINTS	DESCRIPTION
`studentId`	UUID	Yes	—	Student for whom payment is made
`centerId`	UUID	Yes	—	Center receiving the payment
`batchId`	UUID	No	—	Specific batch this payment is for
`amountPaid`	decimal	Yes	Min 0.01	Amount paid
`currency`	string	No	—	Currency code (defaults to center's currency)
`paymentDate`	date	Yes	ISO-8601 (`YYYY-MM-DD`)	Date the payment was made
`referenceNumber`	string	Yes	Non-blank	Bank/gateway transaction reference
`remarks`	string	No	—	Optional notes

Response: `201 Created`

```
{
  "id": "ii3r1733-r5mo-74q7-n049-779988773333",
  "parentId": "gg1p9511-p3km-52o5-1827-557766551111",
  "studentId": "dd8m6288-m0hj-2912-i594-224433228888",
  "centerId": "550e8400-e29b-41d4-a716-446655440000",
```

```
    "batchId": "772g0622-g4bd-63f6-c938-668877662222",
    "amountPaid": 5000.00,
    "currency": "INR",
    "paymentDate": "2026-03-05",
    "referenceNumber": "NEFT20260305XYZ123",
    "remarks": "March fee",
    "status": "RECORDED",
    "createdAt": "2026-03-07T10:00:00Z"
  }
```

Error Responses:

STATUS	CONDITION
400	Validation failure
403	Insufficient access
404	Parent profile, student, or center not found

Example:

```
curl -X POST https://api.edutech.com/api/v1/parents/gg1p9511-p3km-52o5-l827-557766551111/payments
\
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..." \
  -H "Content-Type: application/json" \
  -d '{
    "studentId": "dd8m6288-m0hj-2912-i594-224433228888",
    "centerId": "550e8400-e29b-41d4-a716-446655440000",
    "batchId": "772g0622-g4bd-63f6-c938-668877662222",
    "amountPaid": 5000.00,
    "currency": "INR",
    "paymentDate": "2026-03-05",
    "referenceNumber": "NEFT20260305XYZ123",
    "remarks": "March fee"
  }'
```

GET /api/v1/parents/{profileId}/payments

List all fee payment records for a parent profile.

Auth required: Yes — profile owner or `CENTER_ADMIN` / `SUPER_ADMIN`

Path Parameters:

NAME	TYPE	DESCRIPTION
`profileId`	UUID	Parent profile identifier

Response: `200 OK` — Array of FeePaymentResponse objects.

Example:

```
curl https://api.edutech.com/api/v1/parents/gg1p9511-p3km-52o5-1827-557766551111/payments \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

POST /api/v1/parents/{profileId}/notification-preferences

Create or update a notification preference by channel and event type (upsert semantics).

Auth required: Yes — profile owner or `SUPER_ADMIN`

Path Parameters:

NAME	TYPE	DESCRIPTION
`profileId`	UUID	Parent profile identifier

Request Body:

FIELD	TYPE	REQUIRED	DESCRIPTION
`channel`	string (NotificationChannel enum)	Yes	Delivery channel (e.g., `EMAIL`, `SMS`, `PUSH`)
`eventType`	string	Yes	Event type identifier (e.g., `FEE_DUE`, `ATTENDANCE_MARKED`, `EXAM_RESULT`)
`enabled`	boolean	Yes	Whether notifications for this channel+event are enabled

Response: `200 OK`

```
{
  "id": "jj4s2844-s6np-85r8-o150-880099884444",
  "parentId": "gg1p9511-p3km-52o5-1827-557766551111",
  "channel": "SMS",
  "eventType": "ATTENDANCE_MARKED",
  "enabled": true,
  "createdAt": "2026-03-07T10:00:00Z"
}
```

Error Responses:

STATUS	CONDITION
400	Missing required fields
403	Not the profile owner
404	Profile not found

Example:

```
curl -X POST https://api.edutech.com/api/v1/parents/gg1p9511-p3km-52o5-1827-557766551111/notificat
ion-preferences \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..." \
  -H "Content-Type: application/json" \
  -d '{
    "channel": "SMS",
    "eventType": "ATTENDANCE_MARKED",
    "enabled": true
  }'
```

GET /api/v1/parents/{profileId}/notification-preferences

List all notification preferences for a parent profile.

Auth required: Yes — profile owner or `SUPER_ADMIN`

Path Parameters:

NAME	TYPE	DESCRIPTION
`profileId`	UUID	Parent profile identifier

Response: `200 OK` — Array of NotificationPreferenceResponse objects.

Example:

```
curl https://api.edutech.com/api/v1/parents/gg1p9511-p3km-52o5-1827-557766551111/notification-pref
erences \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

PUT /api/v1/parents/{profileId}/notification-preferences/{prefId}

Update the enabled/disabled state of a specific notification preference.

Auth required: Yes — profile owner or `SUPER_ADMIN`

Path Parameters:

NAME	TYPE	DESCRIPTION
`profileId`	UUID	Parent profile identifier
`prefId`	UUID	Notification preference identifier

Request Body:

FIELD	TYPE	REQUIRED	DESCRIPTION
`enabled`	boolean	Yes	New enabled state

Response: `200 OK` — Updated NotificationPreferenceResponse.

Error Responses:

STATUS	CONDITION
400	Missing enabled field
403	Not the profile owner
404	Preference not found

Example:

```
curl -X PUT https://api.edutech.com/api/v1/parents/gg1p9511-p3km-52o5-1827-557766551111/notificati
on-preferences/jj4s2844-s6np-85r8-o150-880099884444 \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..." \
  -H "Content-Type: application/json" \
  -d '{ "enabled": false }'
```

Assessments — assess-svc

Routes through api-gateway: `/api/v1/exams/**`, `/api/v1/students/**`

All endpoints require `Authorization: Bearer <access_token>`.

POST /api/v1/exams

Create a new exam for a batch.

Auth required: Yes — `TEACHER`, `CENTER_ADMIN`, or `SUPER_ADMIN`

Request Body:

FIELD	TYPE	REQUIRED	CONSTRAINTS	DESCRIPTION
`title`	string	Yes	Non-blank	Exam title
`description`	string	No	—	Exam description
`batchId`	UUID	Yes	—	Batch this exam belongs to
`centerId`	UUID	Yes	—	Center this exam belongs to
`mode`	string (ExamMode enum)	Yes	—	Mode: `ONLINE` or `OFFLINE`
`durationMinutes`	int	Yes	Min 1	Exam duration in minutes
`maxAttempts`	int	Yes	Min 1	Maximum allowed attempts per student
`startAt`	datetime	No	ISO-8601 instant	Exam window open time
`endAt`	datetime	No	ISO-8601 instant	Exam window close time
`totalMarks`	double	Yes	Min 0.01	Total marks available
`passingMarks`	double	Yes	Min 0.01	Marks required to pass

Response: `201 Created`

```
{
  "id": "kk5t3955-t7oq-96s9-p261-991100995555",
  "batchId": "772g0622-g4bd-63f6-c938-668877662222",
  "centerId": "550e8400-e29b-41d4-a716-446655440000",
  "title": "Physics Unit Test — Kinematics",
  "description": "Chapters 1-3 of NCERT Physics Part 1",
  "mode": "ONLINE",
  "durationMinutes": 90,
  "maxAttempts": 1,
  "startAt": "2026-03-10T09:00:00Z",
  "endAt": "2026-03-10T12:00:00Z",
  "totalMarks": 100.0,
  "passingMarks": 40.0,
  "status": "DRAFT",
  "createdAt": "2026-03-07T10:00:00Z"
}
```

Error Responses:

STATUS	CONDITION
400	Validation failure
403	Insufficient role
404	Batch or center not found

Example:

```
curl -X POST https://api.edutech.com/api/v1/exams \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..." \
  -H "Content-Type: application/json" \
  -d '{
    "title": "Physics Unit Test — Kinematics",
    "description": "Chapters 1-3 of NCERT Physics Part 1",
    "batchId": "772g0622-g4bd-63f6-c938-668877662222",
    "centerId": "550e8400-e29b-41d4-a716-446655440000",
    "mode": "ONLINE",
    "durationMinutes": 90,
    "maxAttempts": 1,
    "startAt": "2026-03-10T09:00:00Z",
    "endAt": "2026-03-10T12:00:00Z",
    "totalMarks": 100.0,
    "passingMarks": 40.0
  }'
```

GET /api/v1/exams/{examId}

Get full details of a specific exam.

Auth required: Yes (any role with access to the exam's batch)

Path Parameters:

NAME	TYPE	DESCRIPTION
`examId`	UUID	Exam identifier

Response: `200 OK` — ExamResponse (see POST /exams response schema).

Error Responses:

STATUS	CONDITION
403	No access to this exam's batch
404	Exam not found

Example:

```
curl https://api.edutech.com/api/v1/exams/kk5t3955-t7oq-96s9-p261-991100995555 \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

GET /api/v1/exams

List all exams for a given batch.

Auth required: Yes (any role with access to the batch)

Query Parameters:

NAME	TYPE	REQUIRED	DESCRIPTION
`batchId`	UUID	Yes	Filter exams by batch

Response: `200 OK` — Array of ExamResponse objects.

Example:

```
curl "https://api.edutech.com/api/v1/exams?batchId=772g0622-g4bd-63f6-c938-668877662222" \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

PUT /api/v1/exams/{examId}/publish

Publish an exam, making it visible and accessible to enrolled students. Only DRAFT exams can be published.

Auth required: Yes — `TEACHER`, `CENTER_ADMIN`, or `SUPER_ADMIN`

Path Parameters:

NAME	TYPE	DESCRIPTION
`examId`	UUID	Exam identifier

Request Body: None

Response: `200 OK` — Updated ExamResponse with `status: "PUBLISHED"`.

Error Responses:

STATUS	CONDITION
403	Insufficient role
404	Exam not found

STATUS	CONDITION
422	Exam is not in DRAFT status, or has no questions

Example:

```
curl -X PUT https://api.edutech.com/api/v1/exams/kk5t3955-t7oq-96s9-p261-991100995555/publish \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

POST /api/v1/exams/{examId}/questions

Add a multiple-choice question to an exam. The exam must be in DRAFT status.

Auth required: Yes — `TEACHER`, `CENTER_ADMIN`, or `SUPER_ADMIN`

Path Parameters:

NAME	TYPE	DESCRIPTION
`examId`	UUID	Exam identifier

Request Body:

FIELD	TYPE	REQUIRED	CONSTRAINTS	DESCRIPTION
`questionText`	string	Yes	Non-blank	The question text
`options`	array of string	Yes	Min 2 items	Answer options
`correctAnswer`	int	Yes	Min 0	Zero-based index of the correct option
`explanation`	string	No	—	Explanation of the correct answer
`marks`	double	Yes	Min 0.01	Marks awarded for a correct answer
`difficulty`	double	No	—	IRT difficulty parameter (b-parameter)
`discrimination`	double	No	—	IRT discrimination parameter (a-parameter)
`guessingParam`	double	No	—	IRT guessing parameter (c-parameter)

Response: `201 Created`

```
{
  "id": "ll6u4066-u8pr-07t0-q372-002211006666",
  "examId": "kk5t3955-t7oq-96s9-p261-991100995555",
  "questionText": "A ball is thrown horizontally at 20 m/s. What is the horizontal displacement af
ter 3 seconds?",
  "options": ["40 m", "60 m", "80 m", "100 m"],
  "correctAnswer": 1,
  "explanation": "Horizontal displacement = v × t = 20 × 3 = 60 m",
  "marks": 2.0,
  "difficulty": 0.4,
  "discrimination": 1.2,
  "guessingParam": 0.25,
  "createdAt": "2026-03-07T10:30:00Z"
}
```

Error Responses:

STATUS	CONDITION
400	Fewer than 2 options, or correctAnswer out of range
403	Insufficient role
404	Exam not found
422	Exam is not in DRAFT status

Example:

```
curl -X POST https://api.edutech.com/api/v1/exams/kk5t3955-t7oq-96s9-p261-991100995555/questions \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..." \
  -H "Content-Type: application/json" \
  -d '{
    "questionText": "A ball is thrown horizontally at 20 m/s. What is the horizontal displacement
after 3 seconds?",
    "options": ["40 m", "60 m", "80 m", "100 m"],
    "correctAnswer": 1,
    "explanation": "Horizontal displacement = v × t = 20 × 3 = 60 m",
    "marks": 2.0,
    "difficulty": 0.4,
    "discrimination": 1.2,
    "guessingParam": 0.25
  }'
```

GET /api/v1/exams/{examId}/questions

List all questions for an exam.

Auth required: Yes (any role with access to the exam)

Path Parameters:

NAME	TYPE	DESCRIPTION
`examId`	UUID	Exam identifier

Response: `200 OK` — Array of QuestionResponse objects.

Example:

```
curl https://api.edutech.com/api/v1/exams/kk5t3955-t7oq-96s9-p261-991100995555/questions \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

POST /api/v1/exams/{examId}/enrollments

Enroll a student in an exam.

Auth required: Yes — `TEACHER`, `CENTER_ADMIN`, or `SUPER_ADMIN`

Path Parameters:

NAME	TYPE	DESCRIPTION
`examId`	UUID	Exam identifier

Request Body:

FIELD	TYPE	REQUIRED	DESCRIPTION
`studentId`	UUID	Yes	Auth-svc user ID of the student to enroll

Response: `201 Created`

```
{
  "id": "mm7v5177-v9qs-18u1-r483-113322117777",
  "examId": "kk5t3955-t7oq-96s9-p261-991100995555",
  "studentId": "dd8m6288-m0hj-29l2-i594-224433228888",
  "status": "ENROLLED",
  "enrolledAt": "2026-03-07T11:00:00Z"
}
```

Error Responses:

STATUS	CONDITION
400	Missing studentId
403	Insufficient role
404	Exam or student not found
409	Student already enrolled
422	Exam is not in PUBLISHED status

Example:

```
curl -X POST https://api.edutech.com/api/v1/exams/kk5t3955-t7oq-96s9-p261-991100995555/enrollments
\
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..." \
  -H "Content-Type: application/json" \
  -d '{ "studentId": "dd8m6288-m0hj-29l2-i594-224433228888" }'
```

GET /api/v1/exams/{examId}/enrollments

List all enrollments for an exam.

Auth required: Yes — `TEACHER`, `CENTER_ADMIN`, or `SUPER_ADMIN`

Path Parameters:

NAME	TYPE	DESCRIPTION
`examId`	UUID	Exam identifier

Response: `200 OK` — Array of EnrollmentResponse objects.

Example:

```
curl https://api.edutech.com/api/v1/exams/kk5t3955-t7oq-96s9-p261-991100995555/enrollments \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

POST /api/v1/exams/{examId}/submissions

Start a new exam submission for the authenticated student.

Auth required: Yes — `STUDENT` role

Path Parameters:

NAME	TYPE	DESCRIPTION
`examId`	UUID	Exam identifier

Request Body: None

Response: `201 Created`

```
{
  "id": "nn8w6288-w0rt-29v2-s594-224433228888",
  "examId": "kk5t3955-t7oq-96s9-p261-991100995555",
  "studentId": "dd8m6288-m0hj-2912-i594-224433228888",
  "startedAt": "2026-03-10T09:05:00Z",
  "submittedAt": null,
  "totalMarks": 100.0,
  "scoredMarks": 0.0,
  "percentage": 0.0,
  "status": "IN_PROGRESS",
  "attemptNumber": 1
}
```

Error Responses:

STATUS	CONDITION
403	Caller is not STUDENT or not enrolled
404	Exam not found
422	Exam not yet open, already closed, or max attempts exceeded

Example:

```
curl -X POST https://api.edutech.com/api/v1/exams/kk5t3955-t7oq-96s9-p261-991100995555/submissions
\
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

POST /api/v1/exams/{examId}/submissions/{submissionId}/answers

Submit answers for an in-progress exam submission.

Auth required: Yes — `STUDENT` (submission owner)

Path Parameters:

NAME	TYPE	DESCRIPTION
`examId`	UUID	Exam identifier
`submissionId`	UUID	Submission identifier

Request Body:

FIELD	TYPE	REQUIRED	DESCRIPTION
`answers`	array of AnswerEntry	Yes	One entry per answered question

AnswerEntry object:

FIELD	TYPE	REQUIRED	DESCRIPTION
`questionId`	UUID	Yes	Question identifier
`selectedOption`	int	Yes	Zero-based index of the chosen answer option

Response: `200 OK` — Completed SubmissionResponse with scores calculated.

```
{
  "id": "nn8w6288-w0rt-29v2-s594-224433228888",
  "examId": "kk5t3955-t7oq-96s9-p261-991100995555",
  "studentId": "dd8m6288-m0hj-29l2-i594-224433228888",
  "startedAt": "2026-03-10T09:05:00Z",
  "submittedAt": "2026-03-10T10:30:00Z",
  "totalMarks": 100.0,
  "scoredMarks": 72.0,
  "percentage": 72.0,
  "status": "SUBMITTED",
  "attemptNumber": 1
}
```

Error Responses:

STATUS	CONDITION
400	Empty answers list
403	Not the submission owner
404	Submission or question not found
422	Submission is not in IN_PROGRESS status

Example:

```
curl -X POST https://api.edutech.com/api/v1/exams/kk5t3955-t7oq-96s9-p261-991100995555/submission
s/nn8w6288-w0rt-29v2-s594-224433228888/answers \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..." \
  -H "Content-Type: application/json" \
  -d '{
    "answers": [
      { "questionId": "ll6u4066-u8pr-07t0-q372-002211006666", "selectedOption": 1 },
      { "questionId": "oo9x7399-x1su-30w3-t605-335544339999", "selectedOption": 0 }
    ]
  }'
```

GET /api/v1/exams/{examId}/submissions/{submissionId}

Get the details and current state of a submission.

Auth required: Yes — submission owner, `TEACHER`, `CENTER_ADMIN`, or `SUPER_ADMIN`

Path Parameters:

NAME	TYPE	DESCRIPTION
`examId`	UUID	Exam identifier
`submissionId`	UUID	Submission identifier

Response: `200 OK` — SubmissionResponse.

Error Responses:

STATUS	CONDITION
403	Not the submission owner or insufficient role
404	Submission not found

Example:

```
curl https://api.edutech.com/api/v1/exams/kk5t3955-t7oq-96s9-p261-991100995555/submissions/nn8w628
8-w0rt-29v2-s594-224433228888 \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

GET /api/v1/exams/{examId}/grades

List grade records for all students who submitted the exam.

Auth required: Yes — `TEACHER`, `CENTER_ADMIN`, or `SUPER_ADMIN`

Path Parameters:

NAME	TYPE	DESCRIPTION
`examId`	UUID	Exam identifier

Response: `200 OK`

```
[
  {
    "id": "pp0y8400-y2tv-41x4-u716-446655440000",
    "submissionId": "nn8w6288-w0rt-29v2-s594-224433228888",
    "examId": "kk5t3955-t7oq-96s9-p261-991100995555",
    "studentId": "dd8m6288-m0hj-29l2-i594-224433228888",
    "batchId": "772g0622-g4bd-63f6-c938-668877662222",
    "centerId": "550e8400-e29b-41d4-a716-446655440000",
    "percentage": 72.0,
    "letterGrade": "B+",
    "passed": true,
    "createdAt": "2026-03-10T10:35:00Z"
  }
]
```

Error Responses:

STATUS	CONDITION
403	Insufficient role
404	Exam not found

Example:

```
curl https://api.edutech.com/api/v1/exams/kk5t3955-t7oq-96s9-p261-991100995555/grades \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

GET /api/v1/students/{studentId}/grades

List all grade records for a specific student across all exams.

Auth required: Yes — the student themselves, their linked parent, `TEACHER`, `CENTER_ADMIN`, or `SUPER_ADMIN`

Path Parameters:

NAME	TYPE	DESCRIPTION
`studentId`	UUID	Student's user ID

Response: `200 OK` — Array of GradeResponse objects.

Error Responses:

STATUS	CONDITION
403	No access to this student's records
404	Student not found

Example:

```
curl https://api.edutech.com/api/v1/students/dd8m6288-m0hj-2912-i594-224433228888/grades \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

Psychological Profiling — psych-svc

Routes through api-gateway: `/api/v1/psych/**`

All endpoints require `Authorization: Bearer <access_token>`.

POST /api/v1/psych/profiles

Create a psychometric profile for a student.

Auth required: Yes — `TEACHER`, `CENTER_ADMIN`, or `SUPER_ADMIN`

Request Body:

FIELD	TYPE	REQUIRED	DESCRIPTION
`studentId`	UUID	Yes	Student's user ID
`centerId`	UUID	Yes	Center context
`batchId`	UUID	Yes	Batch context

Response: `201 Created`

```
{
  "id": "qq1z9511-z3uw-52y5-v827-557766551111",
  "studentId": "dd8m6288-m0hj-2912-i594-224433228888",
```

```
    "centerId": "550e8400-e29b-41d4-a716-446655440000",
    "batchId": "772g0622-g4bd-63f6-c938-668877662222",
    "openness": 0.0,
    "conscientiousness": 0.0,
    "extraversion": 0.0,
    "agreeableness": 0.0,
    "neuroticism": 0.0,
    "riasecCode": null,
    "status": "PENDING",
    "createdAt": "2026-03-07T10:00:00Z",
    "updatedAt": "2026-03-07T10:00:00Z"
  }
```

Error Responses:

STATUS	CONDITION
400	Missing required fields
403	Insufficient role
404	Student, center, or batch not found
409	Profile already exists for this student in this batch

Example:

```
curl -X POST https://api.edutech.com/api/v1/psych/profiles \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..." \
  -H "Content-Type: application/json" \
  -d '{
    "studentId": "dd8m6288-m0hj-2912-i594-224433228888",
    "centerId": "550e8400-e29b-41d4-a716-446655440000",
    "batchId": "772g0622-g4bd-63f6-c938-668877662222"
  }'
```

GET /api/v1/psych/profiles/{profileId}

Get a psychometric profile by ID.

Auth required: Yes — `TEACHER`, `CENTER_ADMIN`, `SUPER_ADMIN`, or the student themselves

Path Parameters:

NAME	TYPE	DESCRIPTION
`profileId`	UUID	Psychometric profile identifier

Response: `200 OK` — PsychProfileResponse.

Error Responses:

STATUS	CONDITION
403	No access to this profile
404	Profile not found

Example:

```
curl https://api.edutech.com/api/v1/psych/profiles/qq1z9511-z3uw-52y5-v827-557766551111 \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

GET /api/v1/psych/profiles

List all psychometric profiles for a center.

Auth required: Yes — `CENTER_ADMIN` or `SUPER_ADMIN`

Query Parameters:

NAME	TYPE	REQUIRED	DESCRIPTION
`centerId`	UUID	Yes	Filter profiles by center

Response: `200 OK` — Array of PsychProfileResponse objects.

Example:

```
curl "https://api.edutech.com/api/v1/psych/profiles?centerId=550e8400-e29b-41d4-a716-446655440000"
\
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

POST /api/v1/psych/profiles/{profileId}/sessions

Start a new psychometric assessment session for a student profile.

Auth required: Yes — `TEACHER`, `CENTER_ADMIN`, or `SUPER_ADMIN`

Path Parameters:

NAME	TYPE	DESCRIPTION
`profileId`	UUID	Psychometric profile identifier

Request Body:

FIELD	TYPE	REQUIRED	DESCRIPTION
`sessionType`	string (SessionType enum)	Yes	Session type (e.g., `INITIAL`, `FOLLOW_UP`, `ANNUAL_REVIEW`)
`scheduledAt`	datetime	Yes	ISO-8601 instant — Planned session time

Response: `201 Created`

```
{
  "id": "rr2a0622-a4vx-63z6-w938-668877662222",
  "profileId": "qq1z9511-z3uw-52y5-v827-557766551111",
  "studentId": "dd8m6288-m0hj-2912-i594-224433228888",
  "sessionType": "INITIAL",
  "status": "SCHEDULED",
  "scheduledAt": "2026-03-15T10:00:00Z",
  "startedAt": null,
  "completedAt": null,
  "notes": null,
  "createdAt": "2026-03-07T10:00:00Z"
}
```

Error Responses:

STATUS	CONDITION
400	Missing sessionType or scheduledAt
403	Insufficient role
404	Profile not found

Example:

```
curl -X POST https://api.edutech.com/api/v1/psych/profiles/qq1z9511-z3uw-52y5-v827-557766551111/se
ssions \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..." \
  -H "Content-Type: application/json" \
  -d '{
    "sessionType": "INITIAL",
    "scheduledAt": "2026-03-15T10:00:00Z"
  }'
```

POST /api/v1/psych/profiles/{profileId}/sessions/{sessionId}/complete

Complete a psychometric session by recording the Big Five personality scores and optionally the RIASEC code. This updates the parent profile's trait scores.

Auth required: Yes — `TEACHER`, `CENTER_ADMIN`, or `SUPER_ADMIN`

Path Parameters:

NAME	TYPE	DESCRIPTION
`profileId`	UUID	Psychometric profile identifier
`sessionId`	UUID	Session identifier

Request Body:

FIELD	TYPE	REQUIRED	CONSTRAINTS	DESCRIPTION
`openness`	double	Yes	0.0–1.0	Openness to experience score
`conscientiousness`	double	Yes	0.0–1.0	Conscientiousness score
`extraversion`	double	Yes	0.0–1.0	Extraversion score
`agreeableness`	double	Yes	0.0–1.0	Agreeableness score
`neuroticism`	double	Yes	0.0–1.0	Neuroticism score
`riasecCode`	string	No	—	RIASEC career type code (e.g., `"RIA"`, `"SEC"`)
`notes`	string	No	—	Clinician's session notes

Response: `200 OK` — Updated SessionResponse.

```
{
  "id": "rr2a0622-a4vx-63z6-w938-668877662222",
  "profileId": "qq1z9511-z3uw-52y5-v827-557766551111",
  "studentId": "dd8m6288-m0hj-2912-i594-224433228888",
  "sessionType": "INITIAL",
  "status": "COMPLETED",
  "scheduledAt": "2026-03-15T10:00:00Z",
  "startedAt": "2026-03-15T10:03:00Z",
  "completedAt": "2026-03-15T11:45:00Z",
  "notes": "Student shows strong artistic inclination.",
  "createdAt": "2026-03-07T10:00:00Z"
}
```

Error Responses:

STATUS	CONDITION
400	Trait scores outside 0.0–1.0 range
403	Insufficient role
404	Session or profile not found
422	Session is not in SCHEDULED or IN_PROGRESS status

Example:

```
curl -X POST https://api.edutech.com/api/v1/psych/profiles/qq1z9511-z3uw-52y5-v827-557766551111/se
ssions/rr2a0622-a4vx-63z6-w938-668877662222/complete \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..." \
  -H "Content-Type: application/json" \
  -d '{
    "openness": 0.82,
    "conscientiousness": 0.71,
    "extraversion": 0.45,
    "agreeableness": 0.68,
    "neuroticism": 0.33,
    "riasecCode": "AIR",
    "notes": "Student shows strong artistic inclination."
  }'
```

GET /api/v1/psych/profiles/{profileId}/sessions

List all psychometric sessions for a profile.

Auth required: Yes — `TEACHER`, `CENTER_ADMIN`, `SUPER_ADMIN`, or the student themselves

Path Parameters:

NAME	TYPE	DESCRIPTION
`profileId`	UUID	Psychometric profile identifier

Response: `200 OK` — Array of SessionResponse objects.

Example:

```
curl https://api.edutech.com/api/v1/psych/profiles/qq1z9511-z3uw-52y5-v827-557766551111/sessions \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

POST /api/v1/psych/profiles/{profileId}/career-mappings

Request an AI-generated career mapping for the student's psychometric profile. The profile must have at least one completed session with trait scores.

Auth required: Yes — `TEACHER`, `CENTER_ADMIN`, or `SUPER_ADMIN`

Path Parameters:

NAME	TYPE	DESCRIPTION
`profileId`	UUID	Psychometric profile identifier

Request Body: None

Response: `201 Created`

```
{
  "id": "ss3b1733-b5wy-74a7-x049-779988773333",
  "profileId": "qq1z9511-z3uw-52y5-v827-557766551111",
  "studentId": "dd8m6288-m0hj-2912-i594-224433228888",
  "status": "PENDING",
  "requestedAt": "2026-03-07T10:00:00Z",
  "generatedAt": null,
  "topCareers": null,
  "reasoning": null,
  "modelVersion": null,
  "createdAt": "2026-03-07T10:00:00Z"
}
```

Error Responses:

STATUS	CONDITION
403	Insufficient role
404	Profile not found
422	Profile has no completed sessions with trait data

Example:

```
curl -X POST https://api.edutech.com/api/v1/psych/profiles/qq1z9511-z3uw-52y5-v827-557766551111/career-mappings \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

GET /api/v1/psych/profiles/{profileId}/career-mappings

List all career mapping results for a psychometric profile.

Auth required: Yes — `TEACHER`, `CENTER_ADMIN`, `SUPER_ADMIN`, or the student themselves

Path Parameters:

NAME	TYPE	DESCRIPTION
`profileId`	UUID	Psychometric profile identifier

Response: `200 OK` — Array of CareerMappingResponse objects.

```
[
  {
    "id": "ss3b1733-b5wy-74a7-x049-779988773333",
    "profileId": "qq1z9511-z3uw-52y5-v827-557766551111",
    "studentId": "dd8m6288-m0hj-2912-i594-224433228888",
    "status": "COMPLETED",
    "requestedAt": "2026-03-07T10:00:00Z",
    "generatedAt": "2026-03-07T10:02:15Z",
    "topCareers": "Architect, Industrial Designer, Graphic Designer",
    "reasoning": "High openness and conscientiousness combined with AIR RIASEC suggest careers in
creative technical fields.",
    "modelVersion": "claude-3-opus-20240229",
    "createdAt": "2026-03-07T10:00:00Z"
  }
]
```

Example:

```
curl https://api.edutech.com/api/v1/psych/profiles/qq1z9511-z3uw-52y5-v827-557766551111/career-map
pings \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..."
```

AI Gateway — ai-gateway-svc

Routes through api-gateway: `/api/v1/ai/**`

All endpoints require `Authorization: Bearer <access_token>`. Responses are reactive (non-blocking). An additional `502 Bad Gateway` error is returned when the upstream AI provider is unavailable.

POST /api/v1/ai/completions

Route a chat completion request to the appropriate LLM provider. Provider selection is handled automatically by the gateway based on routing configuration.

Auth required: Yes (any role)

Request Body:

FIELD	TYPE	REQUIRED	DESCRIPTION
`requesterId`	string	Yes	Identifier of the requesting entity (user ID or service name)
`systemPrompt`	string	No	System-level instruction for the LLM
`userMessage`	string	Yes	The user's message or prompt
`maxTokens`	int	Yes	Maximum tokens in the generated response
`temperature`	double	Yes	Sampling temperature (0.0 = deterministic, 1.0 = creative)

Response: `200 OK`

```
{
  "requestId": "req-550e8400-e29b-41d4-a716-446655440000",
  "content": "Kinematics is the branch of mechanics that describes the motion of objects without c
onsidering the forces that cause the motion...",
  "provider": "OPENAI",
  "modelUsed": "gpt-4o",
  "inputTokens": 142,
  "outputTokens": 387,
  "latencyMs": 1243
}
```

`provider` is one of the configured `LlmProvider` enum values (e.g., `OPENAI`, `ANTHROPIC`, `GOOGLE`).

Error Responses:

STATUS	CONDITION
400	Missing required fields
401	Missing or invalid access token
429	Rate limit exceeded
502	Upstream AI provider returned an error or timed out

Example:

```
curl -X POST https://api.edutech.com/api/v1/ai/completions \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..." \
  -H "Content-Type: application/json" \
  -d '{
```

```
    "requesterId": "550e8400-e29b-41d4-a716-446655440000",
    "systemPrompt": "You are a JEE Physics tutor. Provide concise, accurate explanations.",
    "userMessage": "Explain the concept of projectile motion.",
    "maxTokens": 512,
    "temperature": 0.3
  }'
```

POST /api/v1/ai/embeddings

Generate a vector embedding for the provided text using the configured embedding model.

Auth required: Yes (any role)

Request Body:

FIELD	TYPE	REQUIRED	DESCRIPTION
`requesterId`	string	Yes	Identifier of the requesting entity
`text`	string	Yes	Text to embed
`dimensions`	int	Yes	Number of dimensions for the output embedding vector

Response: `200 OK`

```
{
  "requestId": "req-661f9511-f3ac-52e5-b827-557766551111",
  "embedding": [0.0234, -0.1823, 0.5671, ...],
  "modelUsed": "text-embedding-3-small",
  "latencyMs": 88
}
```

Error Responses:

STATUS	CONDITION
400	Missing or empty text, invalid dimensions
401	Missing or invalid access token
429	Rate limit exceeded
502	Upstream embedding provider error

Example:

```
curl -X POST https://api.edutech.com/api/v1/ai/embeddings \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..." \
  -H "Content-Type: application/json" \
  -d '{
    "requesterId": "550e8400-e29b-41d4-a716-446655440000",
    "text": "Projectile motion occurs when an object is launched into the air subject only to grav
itational acceleration.",
    "dimensions": 1536
  }'
```

POST /api/v1/ai/career-predictions

Submit a student's Big Five personality scores and RIASEC code to an AI model for career prediction. Typically called internally by psych-svc after a career mapping is requested, but may also be called directly.

Auth required: Yes — `TEACHER`, `CENTER_ADMIN`, or `SUPER_ADMIN`

Request Body:

FIELD	TYPE	REQUIRED	CONSTRAINTS	DESCRIPTION
`requesterId`	string	Yes	—	Identifier of the requesting entity
`profileId`	string	Yes	—	Psychometric profile ID for correlation
`openness`	double	Yes	0.0–1.0	Big Five openness score
`conscientiousness`	double	Yes	0.0–1.0	Big Five conscientiousness score
`extraversion`	double	Yes	0.0–1.0	Big Five extraversion score
`agreeableness`	double	Yes	0.0–1.0	Big Five agreeableness score
`neuroticism`	double	Yes	0.0–1.0	Big Five neuroticism score
`riasecCode`	string	No	—	RIASEC career interest code (e.g., `"AIR"`)

Response: `200 OK`

```
{
  "requestId": "req-772g0622-g4bd-63f6-c938-668877662222",
  "topCareers": [
    "Architect",
    "Industrial Designer",
    "Urban Planner",
    "Product Designer",
    "Landscape Architect"
```

```
  ],
  "reasoning": "The combination of high openness (0.82) and conscientiousness (0.71) with an AIR R
IASEC profile strongly indicates a preference for structured creative work. Artistic and investiga
tive interests further suggest technical design fields.",
  "modelVersion": "claude-3-opus-20240229",
  "latencyMs": 2187
}
```

Error Responses:

STATUS	CONDITION
400	Missing required fields or trait scores out of range
401	Missing or invalid access token
403	Insufficient role
429	Rate limit exceeded
502	Upstream AI provider error

Example:

```
curl -X POST https://api.edutech.com/api/v1/ai/career-predictions \
  -H "Authorization: Bearer eyJhbGciOiJSUzI1NiJ9..." \
  -H "Content-Type: application/json" \
  -d '{
    "requesterId": "550e8400-e29b-41d4-a716-446655440000",
    "profileId": "qq1z9511-z3uw-52y5-v827-557766551111",
    "openness": 0.82,
    "conscientiousness": 0.71,
    "extraversion": 0.45,
    "agreeableness": 0.68,
    "neuroticism": 0.33,
    "riasecCode": "AIR"
  }'
```

Document generated: 2026-03-07. All field names derived from the Java record source files.